Study Protocol (for reproducibility)

As supplement material to

"Towards Robust and Explainable Depression Detection with Acoustic Foundation Models"

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Master's Graduation Project

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July 11, 2026

Study Protocol

This document describes the full procedure of a two-week mindfulness intervention study as it was eventually run, consolidating the research proposal, lab-visit scripts, consent and debriefing materials, and a pilot study conducted beforehand. Roles are described functionally throughout: an intervention administrator, who explained and led the mindfulness or control practice, and a measurement administrator, who ran the questionnaires, interview, and SART, and who gave instructions pertaining to all other (non-intervention-related) details of the study.

Pilot Study

Before running the full study, the team conducted a small pilot with four participants: two members of the project team and three individuals drawn from the personal network of the project's members, with the first of these not being available for feedback, citing an overflowing schedule. The pilot's purpose was to test the clarity of instructions for the daily practice and voice recordings, the usability of the recording and submission workflow, and the general burden and pacing of the study on participants, and to use this experience to refine the protocol before recruiting the real sample. The pilot ran a shortened version of the design: one week of home practice instead of two, with feedback on the full experience gathered at the single follow-up visit rather than at multiple checkpoints during the week. Pilot participants were not required to meet the PHQ-9 threshold applied to the main sample, and no age restriction was applied to them either. Testing the design on individuals who were not necessarily vulnerable to depression was a deliberate choice, since the team did not want to run an early, less tested version of the protocol on a vulnerable population before the procedure had been checked for basic usability and burden.

Design

The study is a two-arm, between-subjects randomized controlled trial comparing a mindfulness intervention against an active control condition (listening to poetry). Participants are randomly assigned by the measurement administrator to one of the two conditions and remain in that condition for the entire two-week intervention period; they are not informed that a second condition exists, and are only told the comparative purpose of the study at debriefing. The mindfulness intervention drew inspiration from Newman et al. (2026) and mindfulness exercises from Mindfulness-Based Cognitive Therapy (MBCT). Poetry was selected to be publicly available, with a preference for affectively neutral to positive content (compared to other poetry), though this is difficult to measure or control precisely. Participants were not informed that the selection had been made with this criterion in mind.

Toward the end of recruitment, for the final one to five participants, assignment was adjusted manually to balance the number of participants in each condition, since strict random assignment can leave group sizes uneven when the total sample size is not known in advance. The exact software or method used to generate the random assignment itself is not documented here, S. Sheng was responsible for this part of the procedure and can be contacted for the specific mechanism if it is needed for a replication.

Our design required a medical-ethical review to confirm that our study did not fall under Dutch WMO regulations (which concerns medical studies). Randomization was carried out as simple, unstratified assignment for most of the sample.

Recruitment and Eligibility

Recruitment combined two channels: the university's own participant pool, through a study announcement placed there, and physical posters placed around university buildings across different faculties and public buildings in the city, with headers such as “Feeling down lately?”

to invite individuals to participate to “a relaxing practice”, intended to appeal to the target population without stating the study's framing outright. The poster was also distributed digitally through large group chats. Both routes led interested candidates to an online information letter and screening questionnaire.

Eligibility was originally set at ages 18–45, but was narrowed during the study to 18–35, so the achieved sample contains no participants over 35. The PHQ-9 was used as an online screening instrument, with a planned cutoff of 10 or higher, functioning as a heuristic indicator of vulnerability to depression rather than a diagnostic threshold. In practice, a small number of participants scoring 8 or 9 were included, so the realized inclusion criterion was not applied as a strict cutoff. Exclusion criteria were a current psychotic or other major psychiatric disorder, and currently undergoing treatment for any psychiatric problem. No separate screening procedure was used to identify participants in acute mental health crisis; instead, this was addressed entirely through the recruitment and information materials, which discouraged participation by anyone at risk of experiencing severe crises (e.g., suicidal ideation, self-harm, aggression) and directed such individuals elsewhere. The target sample size was 50 to 60 participants meeting the inclusion criteria.

The online screening questionnaire collected, in addition to the PHQ-9, the following fields: consent to be contacted, email address, gender, and education level, along with age as already mentioned above. The screening questionnaire itself is not included as an appendix to this document: it contained the PHQ-9 item text, which is a proprietary instrument and cannot be reproduced here without a license from its publisher.

Overall Timeline

The study spans two in-person lab visits bracketing fourteen days of home practice (totaling sixteen days): lab visit one, then fourteen days of daily practice, then lab visit two. Lab visit two was scheduled within one week of the participant's last day of home practice, so the maximum possible span of the whole study, from first visit to last, was about three weeks, though this was rare in practice. Most participants completed the second visit much closer to the sixteen-day mark.

Facilities

Contrary to an earlier version of the plan involving multiple SART workstations and small groups of participants run together, the study was run one participant at a time, using two adjoining lab rooms. In one room (an eye-tracking lab, though the eye tracker itself was not used for this study) the measurement administrator ran the questionnaires, the SART task, and the interview, and explained every aspect of the study except the intervention itself. In the other room (an EEG lab, with the EEG equipment likewise unused) the intervention administrator explained and facilitated the live mindfulness or poetry practice. Participants waited, seated, outside these two rooms before the session began; there was no separate waiting room.

First Lab Visit

On arrival, the participant is welcomed and given a verbal overview of the study before anything else happens: that it examines how a short relaxation practice done at home relates to well-being, that it involves three sessions (this first visit, at-home practice, and a second visit), and roughly how long each part will take (about two hours for the first visit, ten minutes a day at home for two weeks, about one hour for the second visit). Informed consent is then obtained or re-verified (full text under Consent Form below). The title of the study on the consent form and debriefing letter was: "*Understanding Well-Being and Daily Reflection: An Intervention Study*".

After this, the participant proceeded to the measurement administrator's room for questionnaires, the SART, and the interview.

Questionnaires

Questionnaires are completed on a tablet (iPad). The PHQ-9 is not part of this in-lab battery: it is used only at online screening and again at the follow-up assessment, not administered during the first lab visit itself. The in-lab questionnaire battery instead comprises trait and state measures relevant to the study's outcomes: the Five Facet Mindfulness Questionnaire (FFMQ-15; Gu et al., 2016) for trait mindfulness, the Positive and Negative Affect Schedule (PANAS; Watson et al., 1988), the Brief State Rumination Inventory (BSRI; Marchetti et al., 2018), the State Difficulties in Emotion Regulation Scale (S-DERS; Lavender et al., 2017), and the six-item State-Trait Anxiety Inventory (STAI-6; Marteau & Bekker, 1992). This battery is administered both before and after the live intervention within the same first visit, and again at the second lab visit.

SART Task

The Sustained Attention to Response Task (SART) is administered pre-intervention and post-intervention on the first visit, and again on the second visit. The task uses a word-case go/no-go structure: words appear on screen in either upper or lower case, and the participant is instructed to press a designated key (‘N’) as quickly as possible whenever a word appears in lower case, and to withhold any response when a word appears in upper case. Periodically, the task is interrupted by a thought probe consisting of four questions: what the participant was just thinking about; how self-focused their thoughts were, as opposed to other-focused (with the researcher illustrating the distinction with examples such as “I'm kind” as self-focused versus “the breakfast is delicious” as neutral, and thoughts about another person as other-focused); how positive or negative the thought was; and how difficult it was to disengage from that thought. Responses to these probe questions are given via number keys. A practice block precedes the main task, and the main session begins on the participant pressing the space bar. For further detail on the SART paradigm, its four probe dimensions, and how probe responses and task performance (accuracy, response time) feed into the analysis, see Sheng and van Vugt's pre-registration report, cited in the reference list below.

Interview

The interview was conducted in the same room as the SART and questionnaires, that is, in the measurement administrator's room. Four questions were asked, identical across both lab visits and all three interviews: (1) “How are you feeling today?”; (2) “Can you tell me about a positive event in your life?”; (3) “Can you tell me about a negative event in your life?”; and (4) “What are you looking forward to in the next seven days?”

The first question was answered by the participant recording their own voice message and sending it via the Signal app, mirroring the format used for daily home reflections. The remaining three questions were recorded in the room using a dedicated recorder and a headset microphone worn by the participant: the measurement administrator pressed record as soon as they finished asking each question and pressed stop as soon as the participant finished answering, producing one separate recording per question. Across the two lab visits, this yields nine such recordings per participant in total (three questions × two lab visits, comprising two interviews in the first visit and one interview in the follow-up visit, each recorded as a separate file per question), plus the three Signal-recorded “how are you feeling today” responses, one per interview. Participants were told their answers were for anonymous analysis; there was in the end no active opt-in mechanism letting participants flag individual recordings as listenable versus not, so no recordings were listened to. The instructed minimum response time was softened from “at least one minute” to “about one minute,” to reduce the discomfort some participants appeared to feel when asked to speak continuously.

Recorder operation.

Press HOME to wake the device, press RECORD twice to begin recording, press RECORD once to stop, and press HOME again to save automatically. The battery should be charged beforehand to avoid losing power mid-session.

The Intervention

After the questionnaire and SART blocks, the participant moves to the intervention administrator's room. The intervention administrator first spends some time simply talking with the participant, about the exercises they have just done, the room, or general conversation, before introducing themselves formally and explaining the intervention, probing repeatedly for questions, and then leading the practice live.

The room was set up with comfortable seating, but further environmental touches considered during planning, such as plants, scent, or a neckrest and headrest, were not carried into the final design; comfortable seating was the only environmental element retained. Seating was also positioned as far from the intervention administrator's own seat as the small room allowed (roughly 1.5 meters), so that participants would not feel self-conscious about claiming their own space, while remaining free to sit closer if that felt more comfortable to them.

Depending on the assigned condition, the intervention administrator then explains and leads either the mindfulness practice or the poetry-listening practice. The mindfulness script used for this first live session was largely inspired by the intervention script used by Newman, Cao, Tauber, and van Vugt (2026). No separate written take-home instructions were provided; the

Signal messages sent throughout the intervention period, described below, were considered to serve that function on their own.

Mindfulness Condition.

Participants assigned to the mindfulness condition were told, before the practice began, that the coming two weeks would involve a form of meditation focused on attention: noticing when the mind has become distracted, and returning to a single point of focus in a calm and unforced way. For the first week, that point of focus was the breath alone. From the second week onward, each day's practice still opened with attention on the breath, but then moved to explore one additional point of focus, changing daily. Participants were told this structure was intended to let them first settle into the basic practice before introducing variation.

The framing given to participants was that all instructions, both for this first live session and for the recordings over the following two weeks, were suggestions rather than requirements. Participants were told they were free to set aside anything that did not feel right for them, and that partial engagement with the practice was preferable to pushing through some particular aspect that did not suit them.

The live practice itself began with a brief posture check: participants were invited to sit with hands resting on their knees and spine upright, adjusting until they found a position that felt both relaxed and alert. The intervention administrator told the participant that they too would practice along with the participant throughout. After a few deliberate breaths through the nose, attention was directed to the physical sensation of the breath at the nostrils, noticing where the air felt most distinct on the way in and out.

The practice then moved through three stages. In the first stage, participants counted their breaths silently, one on the in-breath and two on the out-breath, continuing up to ten before starting over, while being told that the breath's pace, depth, and rhythm did not need to be any particular way. Participants were introduced to the practice of noticing when attention had drifted elsewhere, recalling briefly whatever had drawn it away (a sensation, a sound, a thought, a memory), acknowledging it, and returning attention to the breath. In the second stage, counting was replaced with silently noting "in" and "out" in time with the breath. In the third stage, even the silent labeling was dropped, leaving attention on the raw sensation of breathing alone, described to participants using the image of sitting by the sea and watching waves arrive and recede without needing to understand what drives them.

Before ending the practice, participants were asked to notice, without judgment, whether anything had shifted since the start, and were given room to mentally describe to themselves any change if any had occurred in the past ten minutes, or to notice that nothing had changed if that was the case. The session closed with participants slowly returning attention to the room and opening their eyes when ready, followed by an open invitation for any remaining questions before moving on to the next part of the visit.

Poetry Condition.

Participants assigned to the poetry condition were told, before the practice began, that the coming two weeks would involve listening to a selection of written works, read aloud at a slower pace than typical spoken delivery. They were told the selection was made with input from people with more background in written art forms than the intervention administrator themselves, and was deliberately varied to ensure participants would encounter at least some works they would personally enjoy. A different author was featured each day, with all works recited by the intervention administrator.

Participants were told the rationale had two parts: an interest in the effect of simply pausing to sit and listen to these works on their own, and an interest in the combination of that listening with the daily reflection that followed it, in the same way the mindfulness condition paired a practice with a reflection.

Instructions for how to listen were kept informal. Participants were told to find a comfortable position, without any particular posture required, and were free to shift position during listening or to close their eyes if they wished. The emphasis given was on using this moment as a break, taking the listening slowly and receptively. The first live session then proceeded directly into the reading of that day's piece, which will not be included in our documents as it is the only poetry source not publicly available (the author – Philip Larkin – was verbally credited during the intervention session; “Bridge for the Living”, “Oils”, and “A writer” were slowly recited).

Daily Home Practice and the Signal-Based Submission System

The study used Signal's voice-messaging function throughout, both for delivering audio files and for collecting participants' spoken reflections. This removed several downstream considerations from the earlier plan: there was no need, for instance, to instruct participants to cover their phone camera, since no video was recorded and no separate file-upload step to a third-party server was involved.

At the end of the first lab visit, the intervention administrator sets up a private, two-person Signal group chat with the participant, titled “Research Team,” and administered from the intervention administrator's own account. If the participant did not already use Signal, an account was created for them at this point. The chat received the following fixed messages over the course of the study.

Initial welcome message.

Hello! 👋 Welcome to our secure chat. We're excited to start this journey with you!

Today we'll do a quick test so you can see how everything works.

Day 0 test message, with test audio file attached.

🔈🔈🔈🔈🔈🔈Here's the audio file for Day 0. Have a listen, and then send us a voice message (about 1 minute) sharing how you're feeling today.

When you complete the self-guided audio practice, please react to this message with a [checkmark emoji]. If you don't finish it, that's fine and you can also send us a voice message. And you're welcome to continue tomorrow.

Message closing out the lab-visit component and introducing home practice.

Thanks for joining this study, and great job completing the first lab session!

Over the next two weeks, we will send you a daily audio recording to guide your home practice. After you complete each audio practice, please reply to the message with a ✅, and then send a voice message sharing: 'How are you feeling today?' You can speak freely and talk for as long as you want about your feelings or day.

If you complete the full two-week daily practice without missing any days and finish both lab visits, you will receive €50 as compensation! If you miss any days of doing the audio practice, you could also send us a voice message about your daily feelings.

This private chat is used only for sending your daily practice audios and for receiving your voice messages. The chat will be deleted once you complete the study.

Your voice messages will never be listened to individually and will only be used for anonymous data analysis. We will store the data using a unique code so that your personal information remains completely anonymous.

If you have any questions, please feel free to contact us at: s.s.sheng@rug.nl.

For details of this study, please refer to the information letter: [Qualtrics information-letter link (omitted as it is no longer active)].

Daily message, sent each of the fourteen days, with day number and condition-appropriate audio attached.

Here's the audio file for day #. Have a listen, and then record a short voice message (roughly 1 minute, or more) telling us how you're feeling today.

When you complete the audio practice, please react to this message with a ✅. If you don't finish it, that's fine and you can also send us a voice message. And you're welcome to continue tomorrow.

Daily audio files were sent before 8 a.m. each day, which gave the exchange a conversational structure: each party sent roughly one message per day, the participant's audio in the morning and the participant's reflection whenever they completed the practice. No further automated reminder messages beyond these were sent; there was no separate reminder system, and no scheduled mid-study check-in calls were built into the final design, although this had been considered during pilot planning.

Completion was marked by the participant reacting to the day's message with a green checkmark emoji, rather than a thumbs-up as originally considered. The checkmark was chosen deliberately over a thumbs-up, since it requires slightly more effort and was thought to be less prone to being given automatically, without real reflection on whether the practice was actually completed. Reflections were instructed as “about 1 minute, or more,” an open-ended minimum rather than a fixed target, and participants were told they could begin speaking immediately, with no ambient-noise calibration recording requested beforehand. Participants were instructed to hold the phone microphone approximately 2 cm from their mouth for every recording throughout the study, both in the lab and at home.

Participants were told live, during the intervention explanation, that they could contact the research team at any point, including to schedule a video call, if the practice was producing unexpected or unpleasant effects. A separate “who to contact when” information sheet had been considered during planning but was dropped; the live verbal instruction served that purpose instead.

At the end of the fourteen days, once the participant had completed their final home practice, the intervention administrator sent a closing message:

Closing message.

Dear [name], thank you once more for your participation, we greatly appreciate it! We will now leave this chat group and then delete its contents from our account history. On behalf of our research team, we wish you all the best and a good continuation :) –[name of intervention administrator]

The intervention administrator sent this closing message because the intervention administrator would not see the participant again: the second lab visit did not involve the intervention administrator, since it contains no live practice component. The intervention administrator made the participant the admin of the group once the intervention was finished, then left it, then deleted the full chat history from the Research Team account, after first securely storing the participant's voice-message data on a private research virtual workspace.

For the specific content of what participants heard in the daily guided audio recordings themselves, that is, the actual script and delivery of both the mindfulness and poetry conditions across all fourteen days, see the transcripts of the full set of daily audio files (28 in total, 14 per condition). These transcripts are stored in the same location as this study protocol document.

The Signal account used for this purpose was not a personal account belonging to either researcher in daily use. A separate phone and a separate SIM card were used specifically for running the study, so that the account used to contact participants was fully separate from any researcher's personal messaging. Other staff members had this account open on their laptops to check for messages or participants that were missed.

Signal was chosen for its voice messaging function, its ability to fully delete a conversation and its history, and its lack of a commercial data model built on monetizing the platform’s use. A replication in a setting where Signal is not in common use should look for a messaging platform with the same three properties, rather than defaulting to whichever chat app happens to be locally popular, since commercial platforms with different data practices would change the privacy properties described elsewhere in this document.

Second Lab Visit

The second visit runs the same battery as the post-intervention portion of the first visit and takes about an hour. A day-before reminder email had already been sent:

Day-before reminder email.

Dear participant, this email is a reminder that you have scheduled your second lab visit for tomorrow at 9:00. Location: Room 243, Bernoulliborg Building, Zernike Campus, 9747 AG. Important note regarding compensation: to receive your compensation, please register in the UG payment system. Registration requires your IBAN and BSN. You may create your account in advance using the following link: [payment system link]. Please note that your personal information is not accessible to our research team. (If you already have an account, there is no need to register again.) If you have any questions, feel free to contact us. We look forward to seeing you soon.

The visit repeats the same four-question interview as the first visit, with the same recording arrangement (the "how are you feeling today" question via the participant's own phone through Signal, the remaining three questions recorded individually in the room via the dedicated recorder and worn microphone), the same questionnaire battery on the tablet, using the same fixed device passcode noted internally in the lab script, and the same SART task, run from a Mac mini using a Python script within a dedicated conda environment, launched via a fixed set of terminal commands under a shared account (the specific credentials are omitted here). The PHQ-9 is administered again at this visit as a follow-up outcome measure.

The visit closes with debriefing (full text under Debriefing Letter below), followed by compensation.

Compensation

Compensation totals €50 for completing the full two-week intervention and both lab visits, broken down as follows: €16 for the first lab visit; up to €20 for the two weeks of home practice; €8 for the second lab visit; and a €6 completion bonus for finishing without missing any days. Missing five or more days of home practice forfeits the entire €20 home-practice component; missing fewer than five days instead pays €1.50 per completed day. A day counts as completed only if both the audio practice and the accompanying voice reflection were done, as marked by the checkmark reaction described above. Payment itself is submitted through the university's external participant-payment system, which requires the participant's own IBAN and BSN entered directly by them and is not accessible to the research team; the researcher only needs the participant's account identifier, typically their student or personnel number or email, to approve the payment on their end. The €50 total was calculated using a standard rate of €8 per hour, applied to the upper end of the estimated time required for each part of the study. This rate and method can be adjusted for a replication in a different country by substituting a locally appropriate hourly rate and repeating the same calculation against the time estimates given above.

Consent Form (full text)

The consent form, titled "Understanding well-being and daily reflection: an intervention study," is structured as a series of individually checked declarations followed by a signature block.

> *Hereby I, the participant, declare that: I have read and understood the information about the research project and I voluntarily agree to participate. I know whom to contact in case I have questions and I have been informed about my rights. I understand that I can withdraw at any time, without giving a reason, up until and including the last visit to the*

laboratory. I understand that taking part in the study involves the following procedures: (1) the first lab visit (~2 hours): including complete questionnaires, a computerized cognitive task, a brief interview, and a guided audio practice. (2) two-week daily follow-up: 10-minute self-guided audio practice at home, 1-minute (or more) speech recording. (3) the second lab visit (~1 hour): fill out simple questionnaires and cognitive tasks. I understand why the research team needs my personal data and I understand how it is processed and protected. The experimenter has explained the following potential risks to me: potential awareness of unpleasant thoughts/emotions/experiences, fatigue after completing tasks. In addition, I agree: that I will (securely) submit audio recordings of my person as a part of the research project. These raw audio recordings will be destroyed after publication of the research, and until that moment stored on highly secure university servers. That anonymized questionnaire data, behavioral data, and features of speech data that I provided will be deposited on highly secure university servers so that it can be used for future research.

The form closes with fields for date, the participant's full name, and the full name of the researcher present, and a note that the participant has the right to a copy of the form. The withdrawal window as actually run was any point during the study, including up to and during the final lab visit.

Debriefing Letter (full text)

Thank you for participating in this study! Here we say a little more about what the research was really about. Earlier in the information letter, we informed you that the purpose of this research was to study the effects of an at-home intervention on well-being survey responses, performance on a computer-based cognitive task, and features of *speech. We also noted that these measures were to be used to assess potential changes in emotional regulation or cognitive control that may have happened as a result of the intervention. However, we did not disclose the full purpose of the study in order to prevent bias in your responses. The primary purpose of this study was to investigate whether a mindfulness-based intervention would have more beneficial effects on psychological functioning compared to an active control condition. Specifically, we predicted that participants who engaged in mindfulness practice for two weeks would show greater improvements in attentional control (e.g., reduced mind-wandering during the cognitive task), emotional regulation, and speech characteristics (e.g., increased fluency). In addition, we are very interested in how the changes in your way of thinking manifest either in the computerized cognitive task or speech. Confidentiality: although the purpose of the research is not the same as the one you were originally told, all the information you received about how we will treat your data, including your personal data, remains unchanged. Withdrawal: if, after reading this, you want to withdraw from this study and have your data deleted, then please contact the research team. You do not need to explain why you want this, and there will be no negative consequences for you. Contact: we encourage you to share any questions or comments you may have about the study, either by speaking with one of the researchers present or by emailing the research team. Do you have questions or concerns about your rights as a research participant or about the conduct of the research team? In that case, please contact the FSE Research Ethics Committee of the University of Groningen.*

Data Handling

Data collected before the study begins, such as name, email, phone number, and appointment details, is stored in a password-protected file on the university's secure research storage environment (the VRW), accessible only to the three team members running the study, so that scheduling logistics never touch an unprotected or shared medium. Questionnaire responses are collected through Qualtrics and then moved onto the VRW, at which point the copy in Qualtrics is deleted. Voice recordings are stored securely, via the research cloud service described above, until the point of publication: raw audio is deleted once the associated paper is accepted for publication, since the recordings themselves remain a potential re-identification route even after the identity-linking file has been removed, while the other, fully anonymized data types are retained beyond that point for potential future research use, as covered by the informed consent given above. The identity-linking file itself is deleted once data collection is complete; after that point withdrawal is no longer technically possible, since there is no way left to locate a given participant's data to remove it. The specific cloud service used to store voice-message data is not named in this document, for confidentiality and security reasons.

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